1. Semisolid agar layered on the glass slide. A well for antigen and a trough for antiserum cut out of agar.
2. Antigen well filled with human serum.
3. Serum separated by electrophoresis.
4. Antiserum trough filled with antiserum to whole human serum.
5. Serum and antiserum allowed to diffuse into agar.
6. Precipitin lines form for individual serum proteins
    - Counterimmunoelectrophoresis
    - Rocket Electrophoresis

2. Agglutination reactions

When a particulate antigen is mixed with its antibody in the presence of electrolytes at a suitable temperature and pH, the particles are clumped or agglutinated, and the reaction is called agglutination.

ADirect agglutination test

In the direct technique, a cell or insoluble particulate antigen is agglutinated directly by antibody. An example is the agglutination of group A erythrocytes by anti-A sera.

Indirect (Passive) agglutination test

Passive agglutination refers to agglutination of antigen coated cells or inert particles (bentonite or latex particles) which are passive carriers of soluble antigens. An example is the latex agglutination for detection of rheumatoid factor.

- **Applications of agglutination reactions**

a) Slide agglutination

Slide agglutination is a routine test for the identification of many bacterial isolates from clinical specimens. It is also the method used for blood grouping and cross matching.

b) Tube agglutination

This is a standard quantitative method for measurement of antibodies. Widal test done for typhoid and Weil Felix test done for rickettsial infections are examples of Tube agglutination.

Latex agglutination test

Here latex particles are used as passive carriers for adsorbed soluble antigens. The most widespread application of latex agglutination has been in the detection of rheumatoid factor. Latex agglutination tests are also employed in the clinical laboratory for detection of HBs Ag, ASO (Antistreptolysin O) and CRP (Carbohydrate Reactive Protein)

Coombs test (antiglobulin test)

This test was devised by Coombs, Mourant and Race (1945) for the detection of anti-Rh antibodies that do not agglutinate Rh-positive red blood cells in saline. The Coombs test may be of the direct or the indirect type.

Applications of coombs test

1. Erythrocyte typing in blood banks.
2. The evaluation of hemolytic disease of the newborn.
3. The diagnosis of autoimmune hemolytic anemia.

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